

TECHSHIP INTERNSHIP

DAY 8– MongoDB Fundamentals & Database Operations

SUBMITTED BY,

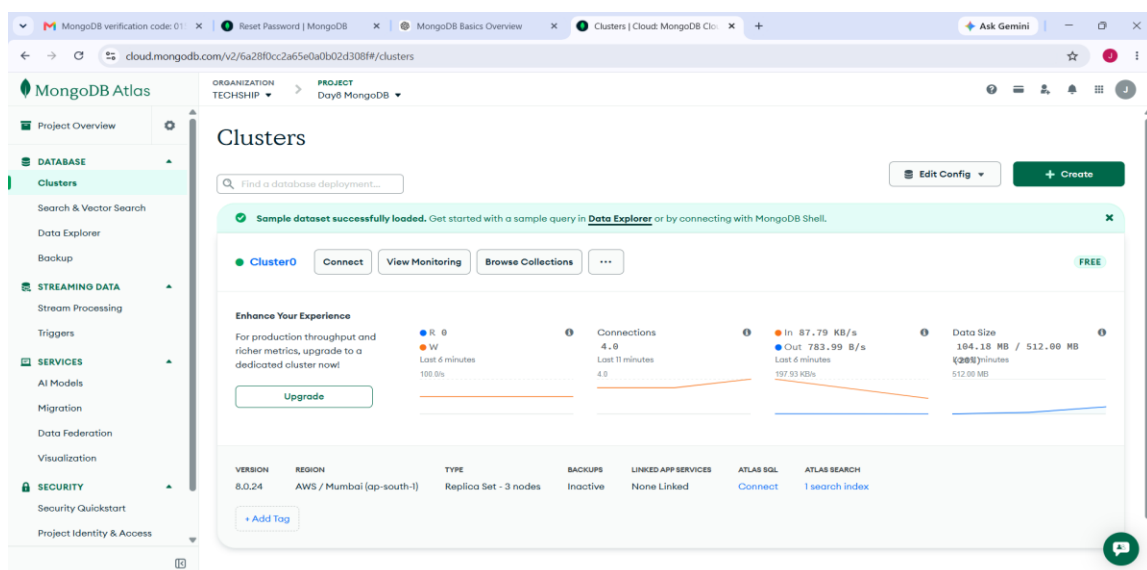
JEEVA JOSHY

10-06-2026

On the eight day of the TECHSHIP Internship Program, I learned the fundamentals of databases and their importance in storing and managing application data. I understood the difference between SQL and NoSQL databases, where SQL uses tables with fixed schemas and MongoDB uses flexible document-based storage. I learned how to use MongoDB Atlas to create clusters, databases, collections, and documents. I also performed CRUD operations such as inserting, reading, updating, and deleting records. The most challenging part was understanding the MongoDB Atlas interface and document structure at the beginning, but after practicing, it became easier. This module helped me understand how modern web applications store and manage data efficiently using MongoDB.

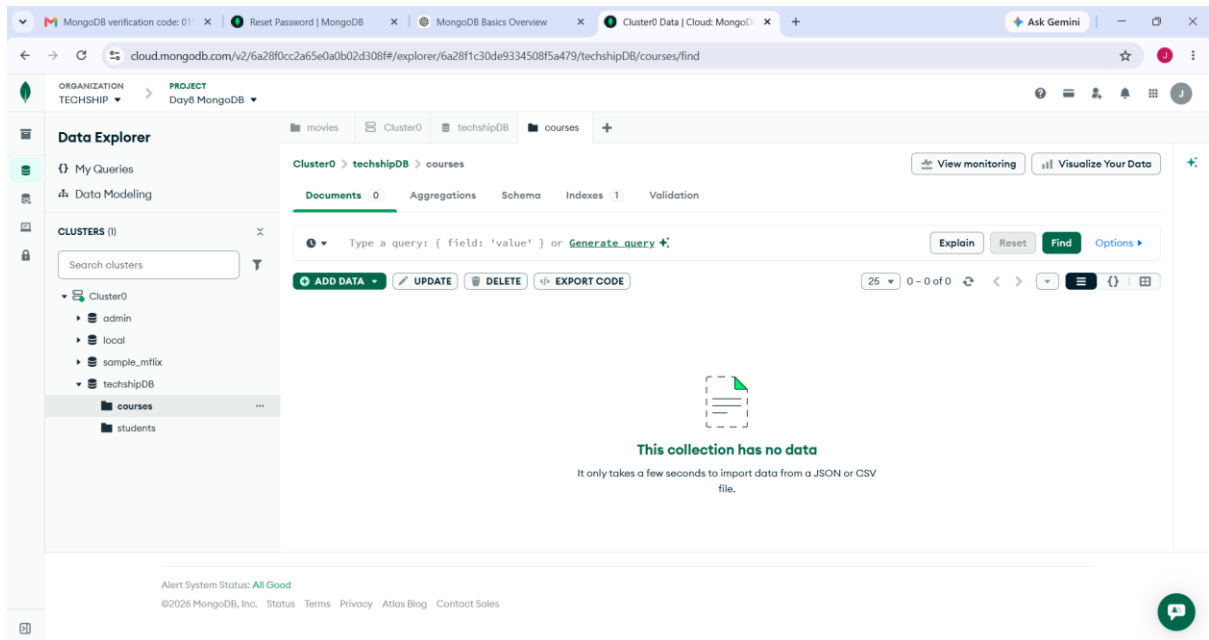
PRACTICAL ACTIVITIES

Activity 1 – Create a MongoDB Atlas Account



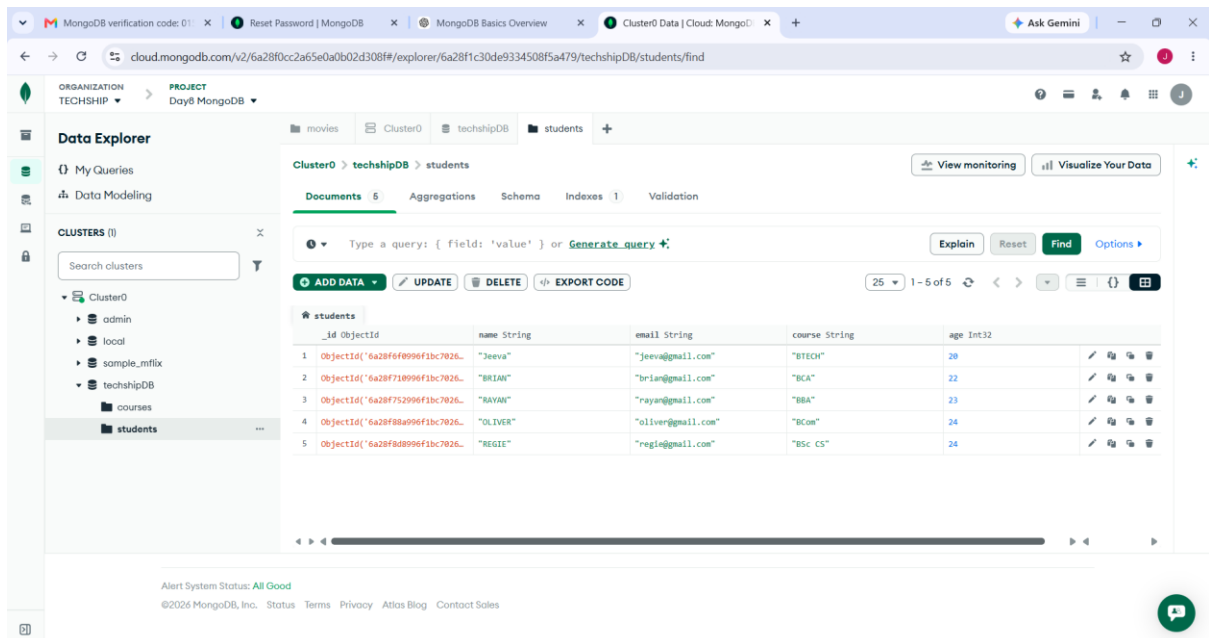
A MongoDB Atlas account was created to access cloud database services.

Activity 2 – Create a Database



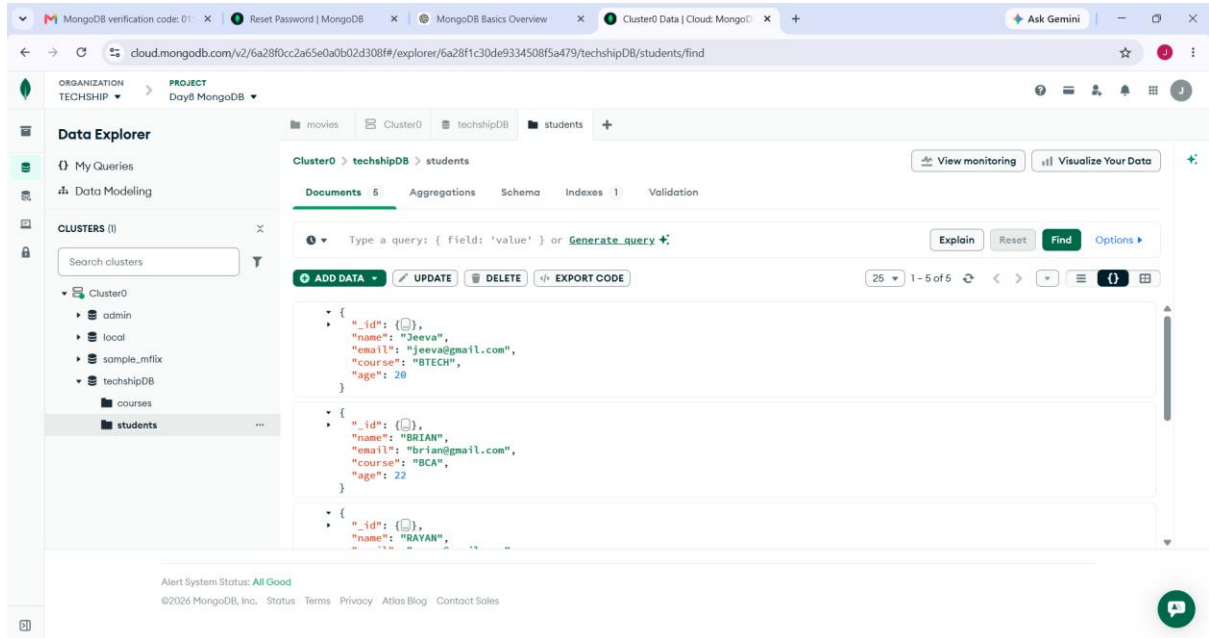
A database was created for storing student-related data.

Activity 3 – Insert Student Records



Student records were inserted into the database.

Activity 4 – Read Data



The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters. The 'students' collection is selected. The main panel shows the 'Documents' tab with a list of documents. The first document is expanded, showing its JSON structure.

Cluster0 > techshipDB > students

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

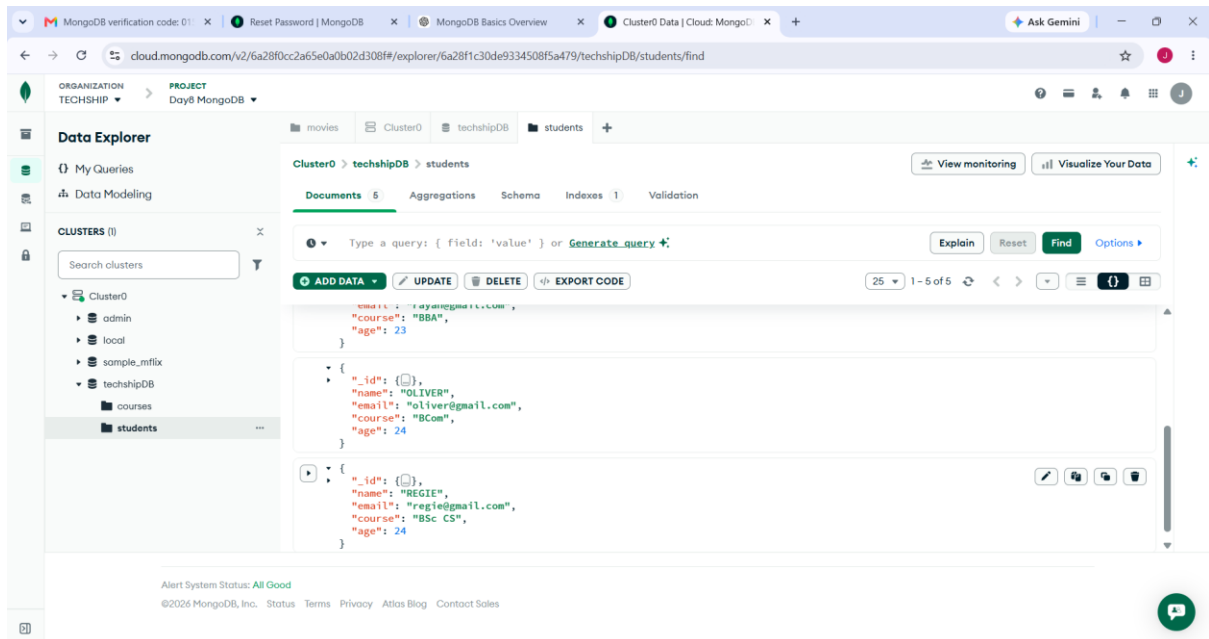
25 1 - 5 of 5

```
{
  "_id": {},
  "name": "Jeeva",
  "email": "jeeva@gmail.com",
  "course": "BTECH",
  "age": 20
}
```

```
{
  "_id": {},
  "name": "BRIAN",
  "email": "brian@gmail.com",
  "course": "BCA",
  "age": 22
}
```

```
{
  "_id": {},
  "name": "RAYAN",
  "email": "rayan@gmail.com",
  "course": "BSC",
  "age": 21
}
```

Alert System Status: All Good
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The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters. The 'students' collection is selected. The main panel shows the 'Documents' tab with a list of documents. The first document is expanded, showing its JSON structure.

Cluster0 > techshipDB > students

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

25 1 - 5 of 5

```
{
  "_id": {},
  "name": "Jeeva",
  "email": "jeeva@gmail.com",
  "course": "BTECH",
  "age": 20
}
```

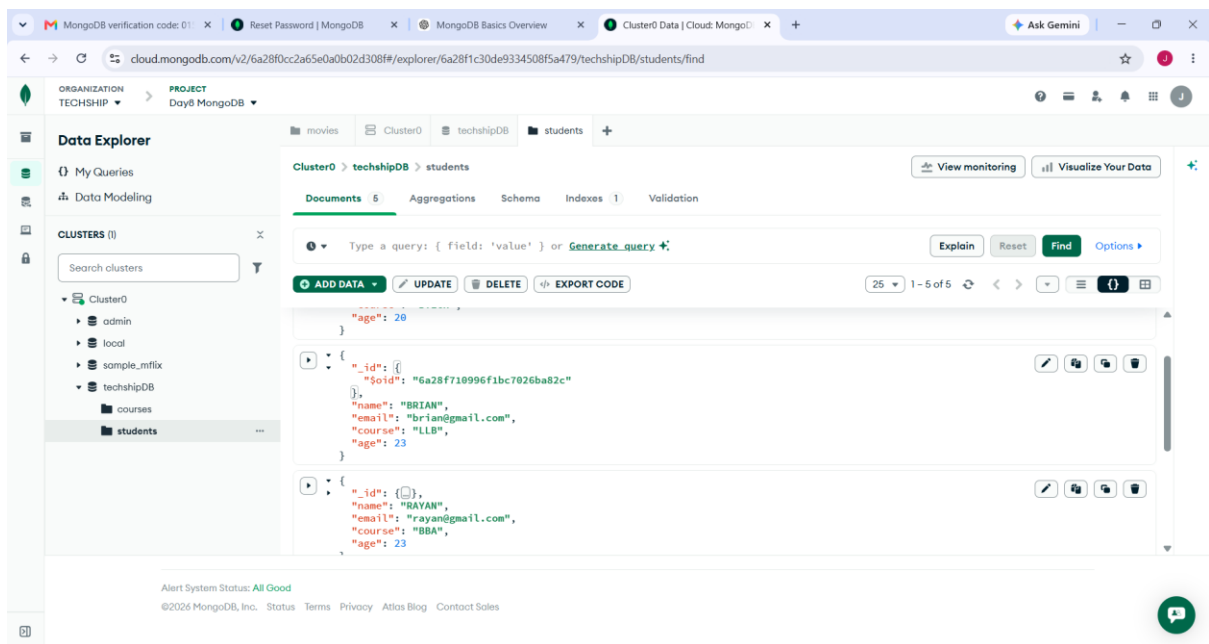
```
{
  "_id": {},
  "name": "BRIAN",
  "email": "brian@gmail.com",
  "course": "BCA",
  "age": 22
}
```

```
{
  "_id": {},
  "name": "RAYAN",
  "email": "rayan@gmail.com",
  "course": "BSC",
  "age": 21
}
```

Alert System Status: All Good
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Stored data was retrieved and viewed from MongoDB.

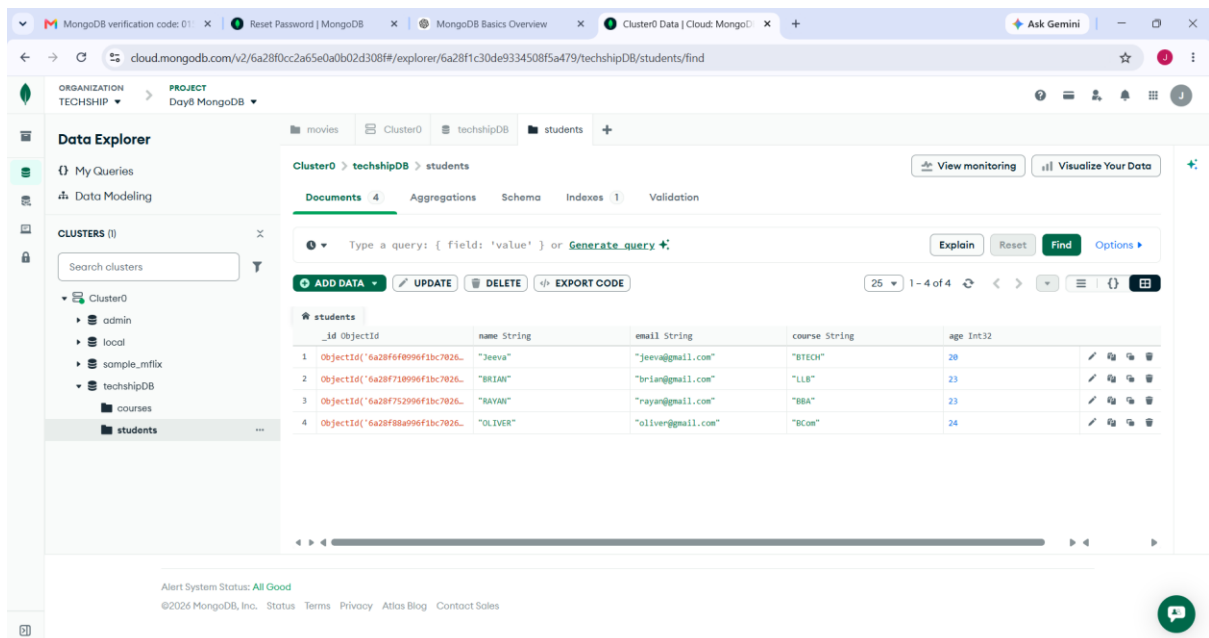
Activity 5 – Update Records



The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'clusters' list with 'Cluster0' selected. The main panel shows the 'students' collection in the 'techshipDB' database. The 'Documents' tab is active, showing a list of documents. The first document is highlighted, showing the following fields: `"_id": {`, `"$oid": "6a28f710996f1bc7026ba82c"`, `"name": "BRIAN"`, `"email": "brian@gmail.com"`, `"course": "LLB"`, and `"age": 23`. The second document is also visible, showing the following fields: `"_id": {`, `"name": "RAYAN"`, `"email": "rayan@gmail.com"`, `"course": "BBA"`, and `"age": 23`. The interface includes a search bar, a query editor, and buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'.

Existing student records were modified.

Activity 6 – Delete Records



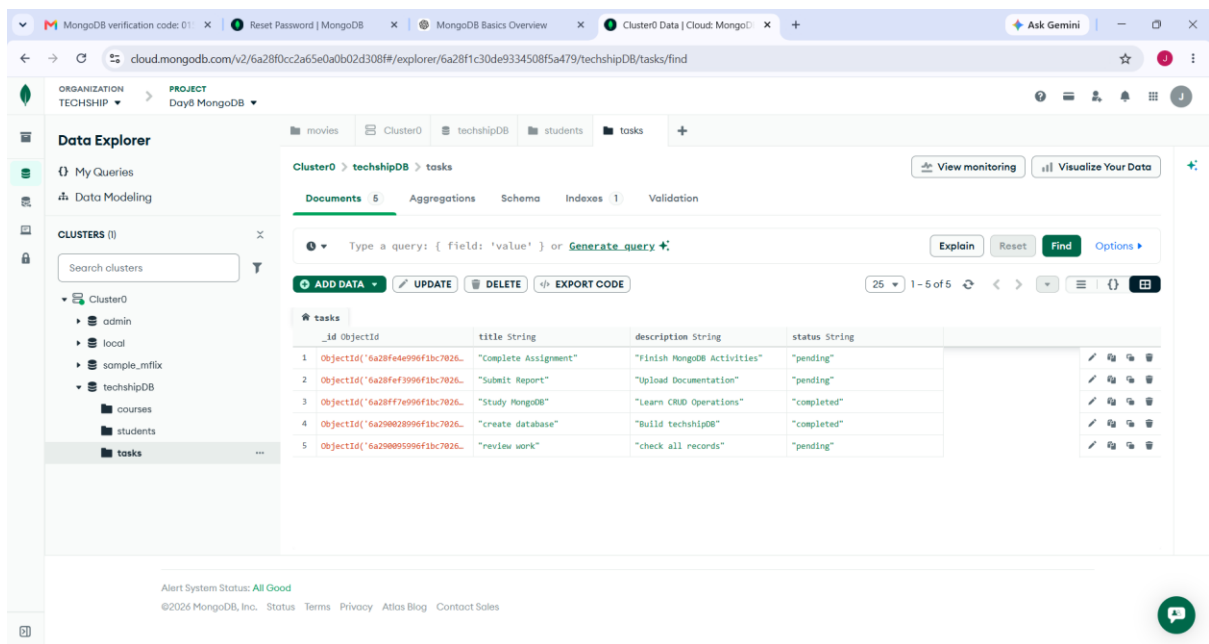
The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'clusters' list with 'Cluster0' selected. The main panel shows the 'students' collection in the 'techshipDB' database. The 'Documents' tab is active, showing a table of documents. The table has the following columns: `_id` (ObjectId), `name` (String), `email` (String), `course` (String), and `age` (Int32). The records are as follows:

	<code>_id</code> ObjectId	<code>name</code> String	<code>email</code> String	<code>course</code> String	<code>age</code> Int32
1	<code>ObjectId('6a28f6f0996f1bc7026...')</code>	Jeeva	<code>"jeeva@gmail.com"</code>	BTECH	20
2	<code>ObjectId('6a28f710996f1bc7026...')</code>	BRIAN	<code>"brian@gmail.com"</code>	LLB	23
3	<code>ObjectId('6a28f752996f1bc7026...')</code>	RAYAN	<code>"rayan@gmail.com"</code>	BBA	23
4	<code>ObjectId('6a28f88a996f1bc7026...')</code>	OLIVER	<code>"oliver@gmail.com"</code>	BCom	24

The interface includes a search bar, a query editor, and buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'.

A student record was removed from the database.

Activity 7– Design a Collection



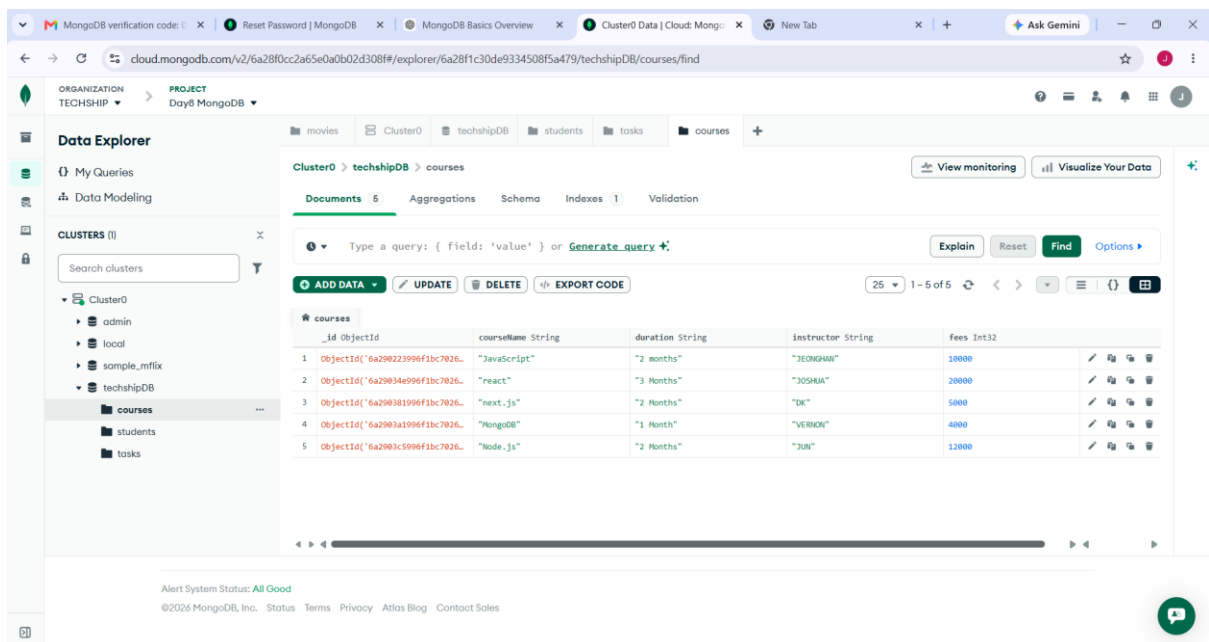
The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters and databases. The main panel shows the 'tasks' collection in the 'techshipDB' database. The collection contains 5 documents, each representing a task with fields: _id, title, description, and status.

#	_id	title	description	status
1	ObjectId('6a28f4e996fbc7026...')	"Complete Assignment"	"Finish MongoDB Activities"	"pending"
2	ObjectId('6a28fef3996fbc7026...')	"Submit Report"	"Upload Documentation"	"pending"
3	ObjectId('6a28ff7e996fbc7026...')	"Study MongoDB"	"Learn CRUD Operations"	"completed"
4	ObjectId('6a290028996fbc7026...')	"create database"	"Build techshipDB"	"completed"
5	ObjectId('6a29005996fbc7026...')	"review work"	"check all records"	"pending"

A task management collection was created for practice.

Challenge Activities

Challenge 1 – Course Management Collection



The screenshot shows the MongoDB Cloud Portal interface. The left sidebar displays the 'Data Explorer' with a tree view of clusters and databases. The main panel shows the 'courses' collection in the 'techshipDB' database. The collection contains 5 documents, each representing a course with fields: _id, courseName, duration, instructor, and fees.

#	_id	courseName	duration	instructor	fees
1	ObjectId('6a290223996fbc7026...')	"JavaScript"	"2 months"	"JEEKSHAN"	10000
2	ObjectId('6a29034e996fbc7026...')	"react"	"3 Months"	"JOSHUA"	20000
3	ObjectId('6a290381996fbc7026...')	"next.js"	"2 Months"	"DK"	5000
4	ObjectId('6a2903a1996fbc7026...')	"MongoDB"	"1 Month"	"VERNON"	4000
5	ObjectId('6a2903c5996fbc7026...')	"Node.js"	"2 Months"	"JUN"	12000

A course management system was created.

Challenge 2 – Employee Database

Create Operations

The screenshot shows the MongoDB Cloud Portal interface. The left sidebar contains 'Data Explorer' and 'CLUSTERS'. The main panel displays the 'employees' collection in the 'techshipDB' database. The 'Documents' tab is active, showing a table of employee records. The table has columns: _id, employeeName, department, salary, and experience. There are three records listed.

	_id	employeeName	department	salary	experience
1	ObjectId('6a290408996f1bc7026...')	"SAANJ"	"IT"	35000	2
2	ObjectId('6a2904ac996f1bc7026...')	"ARJUN"	"HR"	40000	3
3	ObjectId('6a2904c2996f1bc7026...')	"NAYA"	"Finance"	45000	5

An employee management database was created.

Read Operations

This screenshot is identical to the one above, showing the MongoDB Cloud Portal interface with the 'employees' collection in the 'techshipDB' database. The 'Documents' tab is active, displaying a table of employee records with three entries.

	_id	employeeName	department	salary	experience
1	ObjectId('6a290408996f1bc7026...')	"SAANJ"	"IT"	35000	2
2	ObjectId('6a2904ac996f1bc7026...')	"ARJUN"	"HR"	40000	3
3	ObjectId('6a2904c2996f1bc7026...')	"NAYA"	"Finance"	45000	5

Retrieved stored student records from the database.

Update Operation

Cluster0 > hospitalManagementDB > patients

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

```
{
  "_id": {},
  "patientId": "P001",
  "name": "FELIX",
  "email": "felix@gmail.com",
  "age": 24,
  "gender": "Male",
  "phone": "9876543210",
  "address": "MELBOURNE",
  "bloodGroup": "AB"
}
```

```
{
  "_id": {},
  "patientId": "P002",
  "name": "BANG CHAN",
  "email": "gnabnahc@gmail.com",
  "age": 28,
  "gender": "Male",
  "phone": "9876543210"
}
```

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Updated existing student data

Delete Operation

Cluster0 > techshipDB > employees

Documents 2 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

	employees				
	_id ObjectId	employeeName String	department String	salary Int32	experience Int32
1	ObjectId('6a298498996f1bc...')	"SAANJ"	"IT"	30000	2
2	ObjectId('6a2984ac996f1bc...')	"ARJUN"	"HR"	40000	3

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Removed a student record from the collection.

Challenge 3 – Design Your Own Database

Hospital Management Database

The screenshot shows the MongoDB Cloud Portal interface. The left sidebar contains the 'Data Explorer' section with a tree view of clusters and collections. The main panel displays the 'patients' collection in the 'hospitalManagementDB' database. The 'Documents' tab is active, showing two documents. The first document has the following fields: `_id`, `patientId`, `name`, `email`, `age`, `gender`, `phone`, `address`, and `bloodGroup`. The second document has the same fields. The interface includes a search bar, a query editor, and buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The bottom status bar shows 'Alert System Status: All Good' and copyright information for MongoDB, Inc.

The screenshot shows the MongoDB Cloud Portal interface. The left sidebar contains the 'Data Explorer' section with a tree view of clusters and collections. The main panel displays the 'patients' collection in the 'hospitalManagementDB' database. The 'Documents' tab is active, showing two documents. The first document has the following fields: `age`, `gender`, `phone`, `address`, and `bloodGroup`. The second document has the following fields: `_id`, `patientId`, `name`, `email`, `age`, `gender`, `phone`, `address`, and `bloodGroup`. The interface includes a search bar, a query editor, and buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The bottom status bar shows 'Alert System Status: All Good' and copyright information for MongoDB, Inc.

cloud.mongodb.com/v2/6a28f0cc2a65e0a0b02d308f#/explorer/6a28f1c30de9334508f5a479/hospitalManagementDB/Doctors/find

ORGANIZATION TECHSHIP > PROJECT Day8 MongoDB

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Cluster0

admin

hospitalManagementDB

Appointments

Doctors

patients

local

sample_mflix

techshipDB

Cluster0 > hospitalManagementDB > Doctors

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN RESET FIND OPTIONS

ADD DATA UPDATE DELETE EXPORT CODE

25 1 - 3 of 3

```
{
  "_id": {},
  "doctorId": "D002",
  "name": "YEONJUN",
  "specialization": "ONCOLOGYST",
  "experience": 9,
  "phone": "9124356789",
  "email": "yeonjun@gmail.com"
}
```

```
{
  "_id": {},
  "doctorId": "D003",
  "name": "TAEHYUN",
  "specialization": "CARDIOLOGYST",
  "experience": 8,
  "phone": "9123457689",
  "email": "taehyun@gmail.com"
}
```

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cloud.mongodb.com/v2/6a28f0cc2a65e0a0b02d308f#/explorer/6a28f1c30de9334508f5a479/hospitalManagementDB/patients/find

ORGANIZATION TECHSHIP > PROJECT Day8 MongoDB

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Cluster0

admin

hospitalManagem...

patients

local

sample_mflix

techshipDB

Cluster0 > hospitalManagementDB > patients

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN RESET FIND OPTIONS

ADD DATA UPDATE DELETE EXPORT CODE

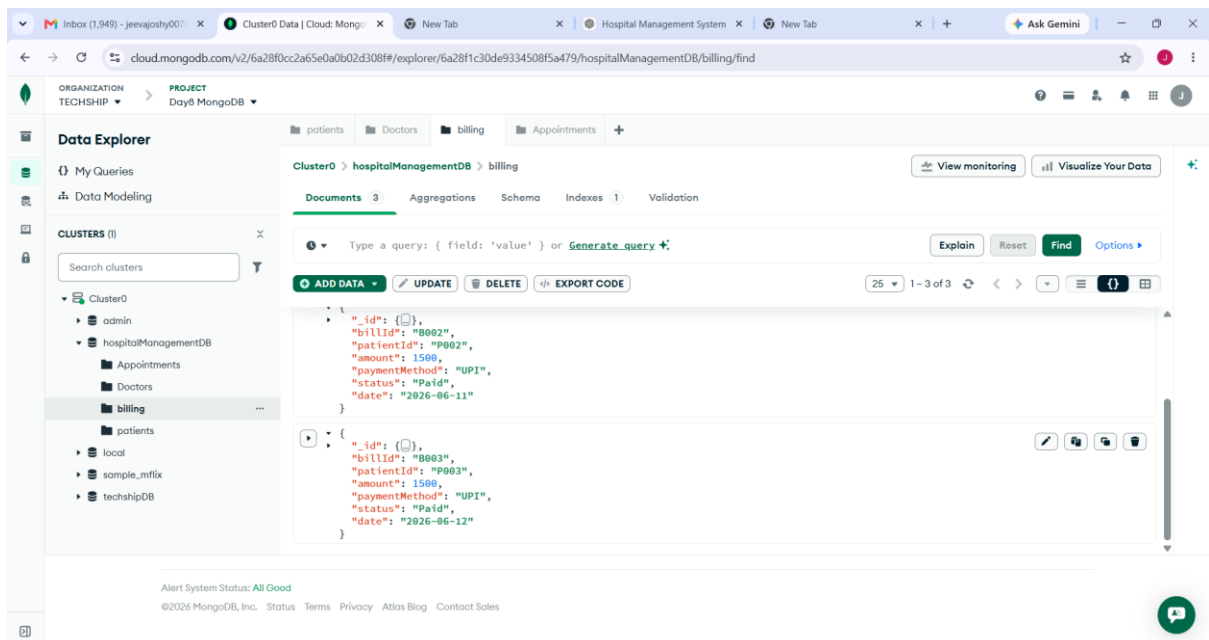
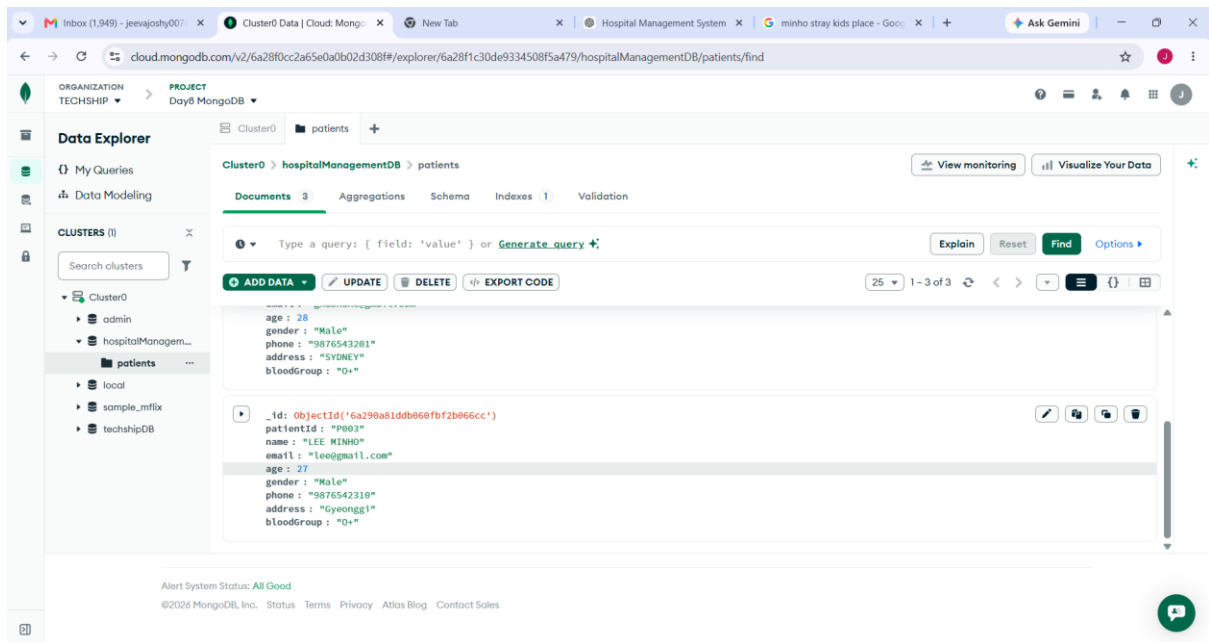
25 1 - 3 of 3

```
_id: ObjectId('6a2988a7ddb060fbf2b066ca')
patientId: "P001"
name: "FELIX"
email: "felix@gmail.com"
age: 25
gender: "Male"
phone: "9876543219"
address: "MELBOURNE"
bloodGroup: "AB"
```

```
_id: ObjectId('6a2989f8ddb060fbf2b066ca')
patientId: "P002"
name: "BANG CHAN"
email: "gnabnahc@gmail.com"
age: 28
gender: "Male"
```

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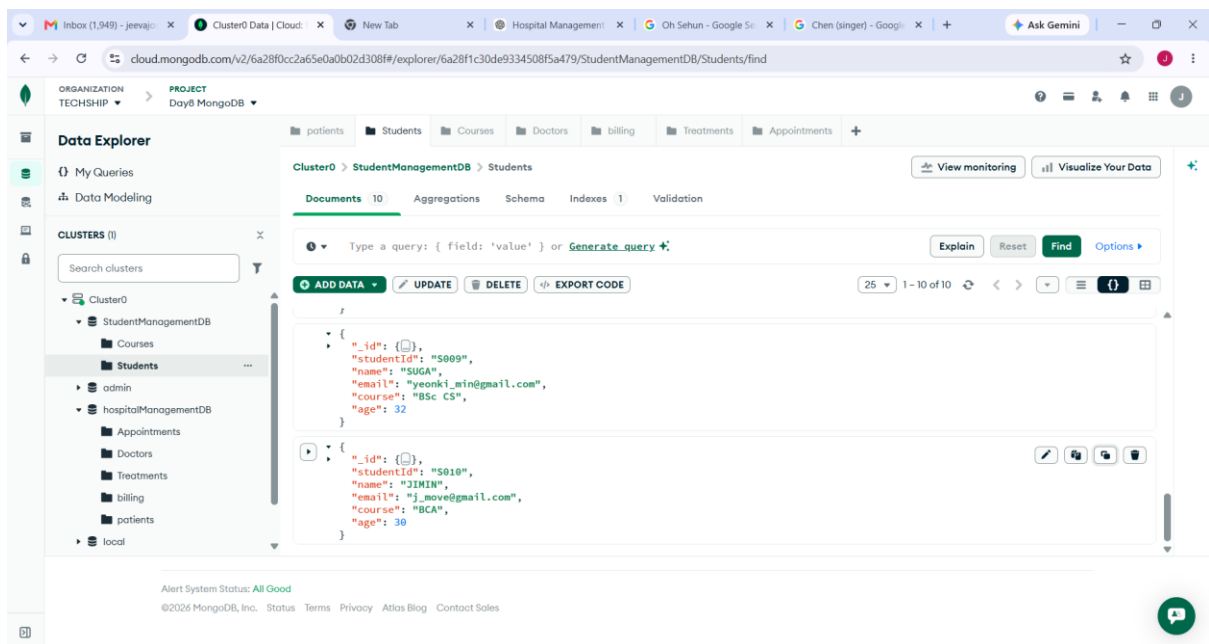
A database named **hospitalManagementDB** was created. Multiple collections, including **patients**, **doctors**, **appointments**, **treatments**, and **billing**, were designed and populated with sample documents. This activity demonstrated how MongoDB can be used to model a real-world hospital management system by efficiently storing and managing healthcare-related data such as patient details, doctor information, appointments, treatments, and billing records.

Assignment

Student Management Database

Designed a complete student management system using MongoDB.

Create Operation



cloud.mongodb.com/v2/6a28f0cc2a65e0a0b02d308f#/explorer/6a28f1c30de9334508f5a479/StudentManagementDB/Students/find

ORGANIZATION TECHSHIP PROJECT Day8 MongoDB

patients Students Courses Doctors billing Treatments Appointments

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - StudentManagementDB
 - Courses
 - Students
 - admin
 - hospitalManagementDB
 - Appointments
 - Doctors
 - Treatments
 - billing
 - patients
 - local

Cluster0 > StudentManagementDB > Students

Documents 10 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

25 1 - 10 of 10

```
{
  "_id": {},
  "studentId": "S001",
  "name": "XIUMIN",
  "email": "xiumin@gmail.com",
  "course": "BCA",
  "age": 36
}
```

```
{
  "_id": {},
  "studentId": "S002",
  "name": "CHEN",
  "email": "chen@gmail.com",
  "course": "BSc CS",
  "age": 33
}
```

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cloud.mongodb.com/v2/6a28f0cc2a65e0a0b02d308f#/explorer/6a28f1c30de9334508f5a479/StudentManagementDB/Students/find

ORGANIZATION TECHSHIP PROJECT Day8 MongoDB

patients Students Courses Doctors billing Treatments Appointments

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - StudentManagementDB
 - Courses
 - Students
 - admin
 - hospitalManagementDB
 - Appointments
 - Doctors
 - Treatments
 - billing
 - patients
 - local

Cluster0 > StudentManagementDB > Students

Documents 10 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

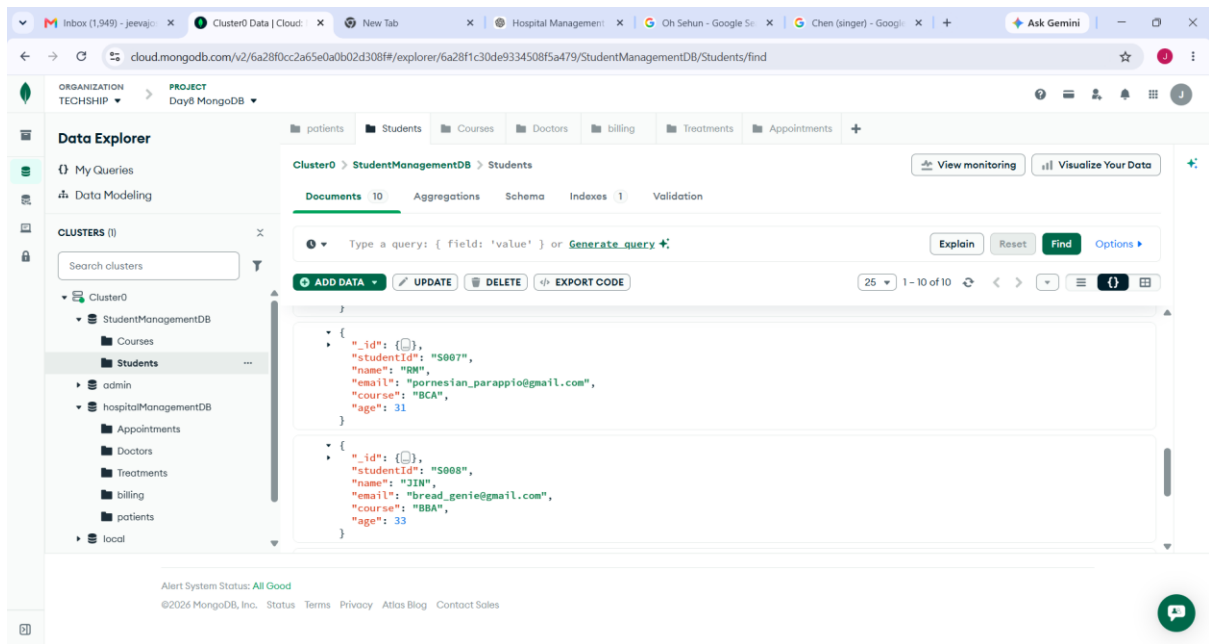
25 1 - 10 of 10

```
{
  "_id": {},
  "studentId": "S005",
  "name": "LUHAN",
  "email": "luhan@gmail.com",
  "course": "BSc Physics",
  "age": 36
}
```

```
{
  "_id": {},
  "studentId": "S006",
  "name": "JUNGKOOK",
  "email": "abcdefghi_lmnopqrstuvwxyz@gmail.com",
  "course": "BCoE",
  "age": 28
}
```

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Successfully created students records.

Read Operation

cloud.mongodb.com/v2/6a28f0cc2a65e0a0b02d308f#/explorer/6a28f1c30de9334508f5a479/StudentManagementDB/Students/find

ORGANIZATION TECHSHIP PROJECT Day8 MongoDB

patients Students Courses Doctors billing Treatments Appointments

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - StudentManagementDB
 - Courses
 - Students
 - admin
 - hospitalManagementDB
 - Appointments
 - Doctors
 - Treatments
 - billing
 - patients
 - local

Cluster0 > StudentManagementDB > Students

Documents 10 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

25 1 - 10 of 10

```
{
  "_id": {},
  "studentId": "S009",
  "name": "SUGA",
  "email": "yeonki_min@gmail.com",
  "course": "BSc CS",
  "age": 32
}
```

```
{
  "_id": {},
  "studentId": "S010",
  "name": "JIMIN",
  "email": "j_move@gmail.com",
  "course": "BCA",
  "age": 30
}
```

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cloud.mongodb.com/v2/6a28f0cc2a65e0a0b02d308f#/explorer/6a28f1c30de9334508f5a479/StudentManagementDB/Students/find

ORGANIZATION TECHSHIP PROJECT Day8 MongoDB

patients Students Courses Doctors billing Treatments Appointments

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - StudentManagementDB
 - Courses
 - Students
 - admin
 - hospitalManagementDB
 - Appointments
 - Doctors
 - Treatments
 - billing
 - patients
 - local

Cluster0 > StudentManagementDB > Students

Documents 10 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

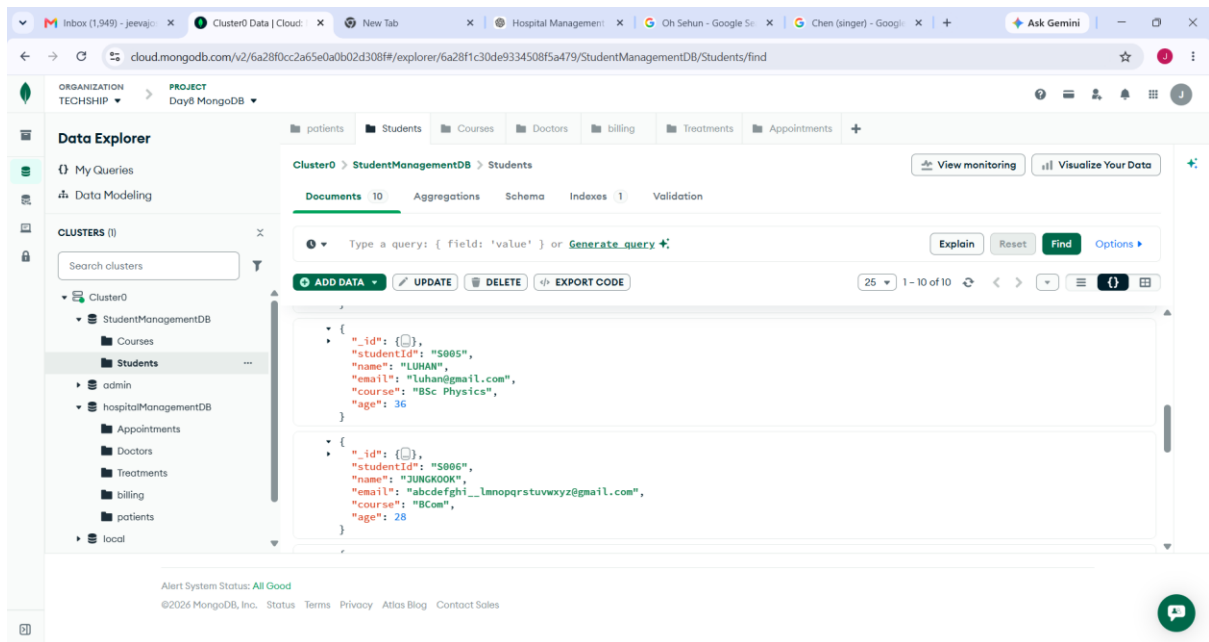
25 1 - 10 of 10

```
{
  "_id": {},
  "studentId": "S001",
  "name": "XIUMIN",
  "email": "xiumin@gmail.com",
  "course": "BCA",
  "age": 36
}
```

```
{
  "_id": {},
  "studentId": "S002",
  "name": "CHEN",
  "email": "chen@gmail.com",
  "course": "BSc CS",
  "age": 33
}
```

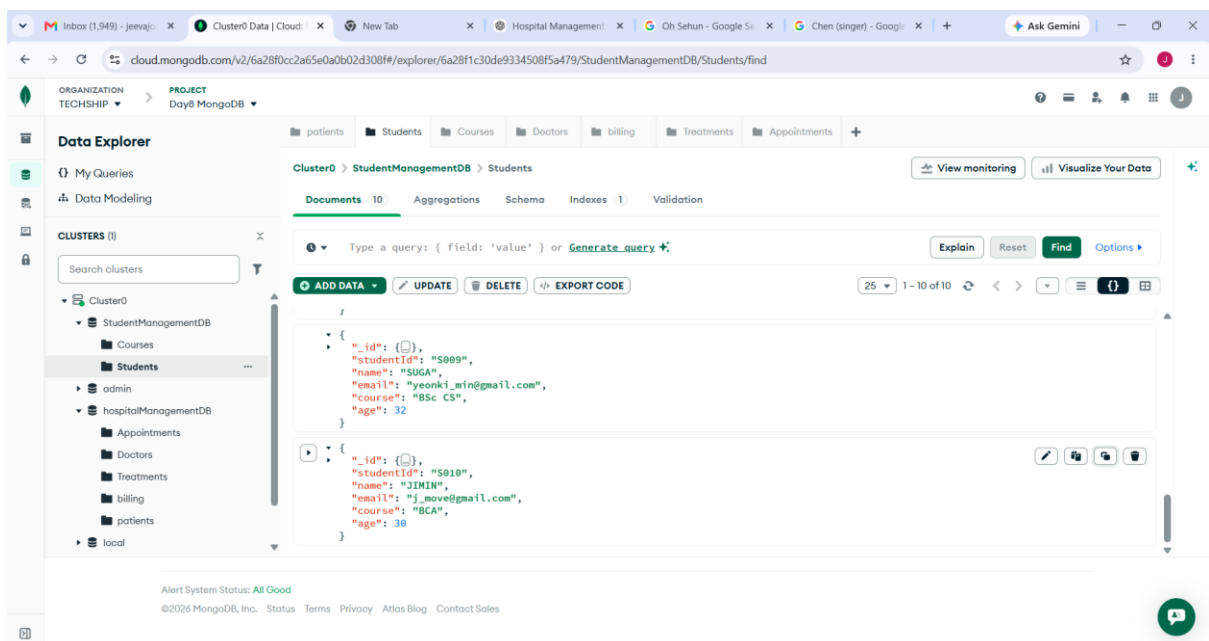
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The Read operation was performed to retrieve and view data stored in the database. This operation helps verify that records have been inserted correctly and allows access to the stored information.

Update Operation



An existing student record was modified using an update query.

Delete Operation

The screenshot displays the MongoDB Cloud console interface. On the left, the 'Data Explorer' sidebar shows a tree view of the database structure, including 'Cluster0', 'StudentManagementDB', and the 'Students' collection. The main panel shows the 'Students' collection with a table of documents. The table has columns: '_id', 'studentId', 'name', 'email', and 'course'. The first document is highlighted, showing its details. Below the table, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. The 'DELETE' button is highlighted, indicating the operation being performed. The bottom status bar shows 'Alert System Status: All Good' and '©2026 MongoDB, Inc.'.

_id	studentId	name	email	course
ObjectId('6a2921802b7c87a...')	"S001"	"XIUMIN"	"xiumin@gmail.com"	"BCA"
ObjectId('6a2921802b7c87a...')	"S002"	"CHEN"	"chen@gmail.com"	"BSc CS"
ObjectId('6a2921802b7c87a...')	"S003"	"SEHUN"	"sehun@gmail.com"	"BBA"
ObjectId('6a2921802b7c87a...')	"S004"	"D.O"	"d_o@gmail.com"	"BCA"
ObjectId('6a2921802b7c87a...')	"S006"	"JUNGKOOK"	"abcdefghi_lmnopqrstuvwxyz"	"BCoE"
ObjectId('6a2921802b7c87a...')	"S007"	"RM"	"pornesian_parappio@gmail.com"	"BCA"
ObjectId('6a2921802b7c87a...')	"S008"	"JIN"	"bread_genie@gmail.com"	"BBA"
ObjectId('6a2921802b7c87a...')	"S009"	"SUGA"	"yeonki_min@gmail.com"	"BSc CS"
ObjectId('6a2921802b7c87a...')	"S010"	"JIMIN"	"i_move@gmail.com"	"BCA"

A student record was removed from the collection using a delete query.